

GUNNEDAH, Australia

WMO: 957400

Lat: 30.9536S

Lon: 150.2494E

Elev: 864

StdP: 14.24

Time Zone: 10.00 (AUS)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.7	31.9	19.9	15.4	59.2	23.3	18.1	55.6	20.7	60.3	18.4	59.2	2.2	120	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.8	100.5	67.7	97.5	67.0	94.5	66.3	72.2	87.7	71.0	86.3	69.8	84.9	11.2	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.4	107.9	76.0	66.9	102.2	75.5	65.4	96.9	75.0	36.5	87.8	35.4	86.6	34.3	84.6	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.1	19.4	17.3	DB	24.4	105.3	2.0	4.3	23.0	108.3	21.8	110.8	20.7	113.2	19.3	116.3
			WB	23.2	75.0	1.7	1.5	22.0	76.1	21.0	76.9	20.0	77.8	18.8	78.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.5	79.3	77.4	72.8	65.4	56.6	51.5	49.4	51.4	57.7	65.4	71.6	75.8	(6)
	DBStd	11.74	5.59	5.59	4.71	5.27	5.09	4.68	4.54	4.97	5.49	6.07	6.16	5.70	(7)
	HDD50	157	0	0	0	0	8	39	67	40	3	0	0	0	(8)
	HDD65	1946	0	1	4	57	261	405	484	422	227	70	13	2	(9)
	CDD50	5436	909	767	708	462	214	84	48	84	235	478	648	799	(10)
	CDD65	1751	444	348	247	69	2	0	0	1	9	83	211	337	(11)
	CDH74	21075	5387	3994	2460	742	42	0	0	18	305	1275	2682	4170	(12)
	CDH80	10179	2987	2077	991	159	1	0	0	2	73	449	1255	2185	(13)
(14) Wind	WSAvg	7.9	9.2	9.6	8.9	8.1	6.4	6.7	6.1	6.7	7.4	8.3	8.9	9.1	(14)
(15) Precipitation	PrecAvg	26.4	4.0	2.6	1.8	1.6	2.2	1.4	1.8	1.6	1.5	2.3	2.6	3.0	(15)
	PrecMax	35.1	11.9	8.3	5.4	5.9	6.7	4.3	6.7	4.0	4.8	5.7	7.4	7.3	(16)
	PrecMin	13.4	0.1	0.1	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.4	0.1	0.0	(17)
	PrecStd	5.6	3.1	2.0	1.4	1.7	1.8	1.1	1.5	1.0	1.1	1.3	1.9	1.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	104.0	104.2	95.9	88.4	78.0	69.5	70.7	76.2	86.7	93.5	101.9	102.6	(19)
		MCWB	68.6	69.6	67.3	62.5	59.0	56.8	54.4	56.5	58.6	61.4	66.8	66.5	(20)
	2%	DB	100.3	99.0	91.8	84.3	74.8	67.0	66.8	71.7	81.6	88.8	95.6	98.4	(21)
		MCWB	68.2	68.3	65.6	62.8	57.7	55.3	54.1	54.7	57.4	61.4	64.4	65.7	(22)
	5%	DB	97.2	95.2	88.7	81.2	72.2	64.9	64.0	68.3	77.1	85.1	91.3	94.8	(23)
		MCWB	67.8	67.7	65.2	61.2	56.6	54.0	52.6	53.1	56.3	60.6	63.6	65.3	(24)
	10%	DB	94.0	91.5	85.7	78.4	69.5	62.5	61.2	65.2	73.1	81.3	87.2	91.4	(25)
		MCWB	67.4	67.1	64.5	60.2	55.6	52.7	51.3	51.7	55.5	59.8	62.7	64.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.0	74.6	71.7	68.1	63.3	59.9	60.7	61.2	63.6	67.2	70.2	71.9	(27)
		MCDB	89.5	91.5	84.8	76.8	70.0	65.0	66.1	71.1	75.6	80.6	87.8	87.4	(28)
	2%	WB	72.4	72.3	69.9	65.5	61.0	57.5	56.9	57.6	61.4	64.9	68.3	70.2	(29)
		MCDB	87.6	88.6	82.7	76.5	69.9	63.1	62.0	66.3	72.5	79.0	83.2	85.7	(30)
	5%	WB	71.1	70.8	68.4	63.5	58.9	55.8	54.3	55.2	59.3	63.3	66.9	69.0	(31)
		MCDB	86.6	86.6	80.7	75.4	68.0	61.9	60.8	64.6	71.1	77.7	81.3	84.2	(32)
	10%	WB	70.0	69.3	66.9	62.0	57.0	54.3	52.3	53.1	57.4	61.8	65.6	67.7	(33)
		MCDB	85.3	84.8	79.6	74.5	67.2	60.5	59.6	63.1	69.6	76.6	80.4	82.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	25.8	25.3	24.2	26.0	26.6	22.9	25.1	28.1	29.5	28.6	27.3	26.9	(35)
		MCDBR	29.9	30.6	28.4	28.5	28.7	25.1	28.3	32.8	36.4	34.3	33.5	32.5	(36)
	5% WB	MCWBR	8.4	8.5	9.3	10.9	14.1	14.5	16.8	18.1	16.5	13.2	10.6	10.1	(37)
		MCWBR	24.1	25.0	22.2	23.1	23.2	20.3	22.1	25.3	27.4	27.1	25.1	24.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.357	0.357	0.331	0.312	0.292	0.287	0.282	0.293	0.325	0.341	0.364	0.357	(40)
		taud	2.527	2.546	2.605	2.619	2.623	2.645	2.642	2.585	2.482	2.473	2.460	2.508	(41)
	Ebn at Noon		312	306	304	294	285	279	286	296	300	307	307	312	(42)
		Edh at Noon	36	34	30	26	23	22	23	27	33	36	38	36	(43)
(44) All-Sky Solar Radiation	RadAvg		2323	2101	1774	1455	1130	897	1025	1324	1723	2021	2176	2330	(44)
	RadStd		146	161	128	86	69	75	64	83	121	167	219	168	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (14 neighbors)	+1.04	N/A	N/A	+2.25	N/A	N/A	-29	N/A	+376	+230	(47)

Nomenclature: See separate page